MIL-I-8580B
2 April 1965
SUPERSEDING
MIL-I-8580A(ASG)
24 October 1955

MILITARY SPECIFICATION

INDICATOR, TACHOMETER, HELICOPTER, 0-4000 RPM, TYPE MA-1

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification covers design and all performance requirements for procurement of a helicopter tachometer indicator.

1.2 Classification - The tachometer indicator shall be of one type having a dual pointer, ratio 1 to 1, to indicate engine rpm and apparent rotor rpm.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

BB-N-411	Nitrogen
DD-G-451	Glass, Flat and Corrugated, for Glazing, Mirrors, and other uses
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Coating, Electrodeposited, Requirements for

Military

MIL-P-116	Preservation, Methods of

FSC 6620

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Military (Continued)

MIL-M-3171	Magnesium Alloy, Process for Corrosion Protection of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-G-5413	Generator, Tachometer, Four Pole
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-R-22256	Reliability Requirements for Design of Electronic Equipment or Systems
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for
MIL-P-23408	Plating: Tin-Cadmium (Electrodeposited)
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-595	Colors

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Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33586	Metals, Definition of Dissimilar
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction - The indicators furnished under this specification shall be a product, which has been tested and meets the Quality Assurance Provisions specified herein.

3.2 Selection of Government documents - Except as provided in 3.2.1 and 3.2.2, specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2.1 Commercial parts - Commercial parts having suitable properties shall be used, when on the date of invitations for bids, there are no suitable standard parts. In any case, commercial parts, such as screws, bolts, nuts, cotter pins, having suitable properties, may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and if practicable, on the contractor's drawings.

3.2.2 Standard parts - With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

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3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Critical materials - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Nonmagnetic materials - Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.3 Metals - Metals shall be of the corrosion resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels, salt spray, or atmospheric conditions to which the indicator may be subjected when in storage or during normal service life.

3.3.3.1 Dissimilar metals - Dissimilar metals as defined in MS33586 shall not be used in intimate contact with each other, unless protection against electrolytic corrosion is provided.

3.3.3.2 Magnesium alloy parts - Magnesium alloy parts shall be treated in accordance with MIL-M-3171. When abrasion resistance is a factor, an anodic treatment approved by the procuring activity shall be used.

3.3.3.3 Aluminum alloy parts - Unless otherwise specified, aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Small holes, pipe threads, and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.3.3.4 Iron and steel parts - Iron and steel parts, exclusive of stator laminations, which are in hermetically sealed cases shall be cadmium plated in accordance with QQ-P-416. Parts which are not in sealed cases shall be chromium, nickel, or zinc plated in accordance with QQ-C-320, QQ-N-290, or QQ-Z-325 respectively. Parts in a confined space in the presence of organic material shall be tin-cadmium plated in accordance with MIL-P-23408. The class and type of plating shall meet the requirements of 3.3.4. Stator laminations shall conform to 3.3.4.

3.3.4 Protective treatment - When materials are used in the construction of the indicator that are subject to atmospheric or environmental conditions likely to cause corrosion in normal service life, they shall be protected against corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification.

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Finishes and protective coatings which will crack, chip, or scale during normal service life or are affected by extremes of atmospheric or environmental conditions, shall not be used.

3.3.5 Fungus-proof materials - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used, they shall be treated with a fungicidal agent acceptable to the procuring activity. If the materials are used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.6 Fumes and vapors - Materials used in the construction of the indicator shall not produce corrosive, deleterious, or toxic fumes or vapors under the conditions specified herein.

3.4 Design and construction - The indicator shall consist of two movements each having a suitable drag-type indicating mechanism driven by a self-starting three-phase synchronous motor, and designed for dual indication and for operation with tachometer generators conforming to MIL-G-5413. It shall be possible to operate both movements from one generator without affecting the operation of either movement. The rotor and engine movements in each indicator shall be similarly identical. The indicator shall be so constructed that no parts will work loose in service, and shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 Maintenance - The design shall be such as to facilitate as much as possible disassembly, repair or overhaul, service maintenance, and reassembly using those tools and items of maintenance equipment which are normally available as commercial standards.

3.4.2 Case - The case shall conform to MS33639, for the three inch size. The length of the case shall be 4 1/2 ±0.5 inches. The case and mounting flange shall be made of nonferrous material, uniform in texture, having a smooth surface, and shall be hermetically sealed. The case shall be finished with a lusterless black material, Color No. 37038 of FED-STD-595. The finishing material shall be of a durable type to withstand usage encountered in service.

3.4.2.1 Hermetic sealing - The case shall provide a hermetically sealed enclosure for all of the mechanism. The case shall be so constructed that it may be opened, the mechanism removed, replaced, and the case resealed at least three times. This shall be possible without the use of any special tool, jig or fixture, unless such device is specifically approved by the procuring activity. The sealing of the case shall not be dependent upon any material which will be adversely affected by any environment to which the indicator may be subjected in normal use in military aircraft.

3.4.2.2 Filling medium - The filling medium shall be a mixture of 90 percent nitrogen and ten percent helium. The nitrogen used shall be in accordance with BB-N-411, Type I, Class I, Grade C. The filling medium shall contain not more than 0.006 milligram of water vapor

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per liter (dew point -65° C) at the filling pressure. The absolute pressure of the filling medium in the case shall be 1 ±0.1 atmosphere.

3.4.3 Weight - The weight of the completely assembled indicator shall not exceed 1.8 pounds.

3.5 Electrical requirements -

3.5.1 Electronic parts - Electronic parts and the application thereof shall be in accordance with MIL-E-5400. Parts that do not appear on approved lists in accordance with MIL-E-5400 shall not be used unless approved by the procuring activity.

3.5.2 Electric connectors - The electric connectors used shall conform to MS33678-10SL-3P except that they shall be hermetically sealed.

3.5.3 Wiring - The three leads from each unit of the indicator winding shall be designated as B, A, and C (conforming to phases 1, 2, and 3). The leads shall be connected to the B, A, and C pins of each electrical receptacle in such manner that when three-phase rotation sequence of B-A-C is applied, the resultant indicator reading is positive. The wiring shall be insulated from the instrument case. The external connections between indicators and generators shall conform to figure 1.

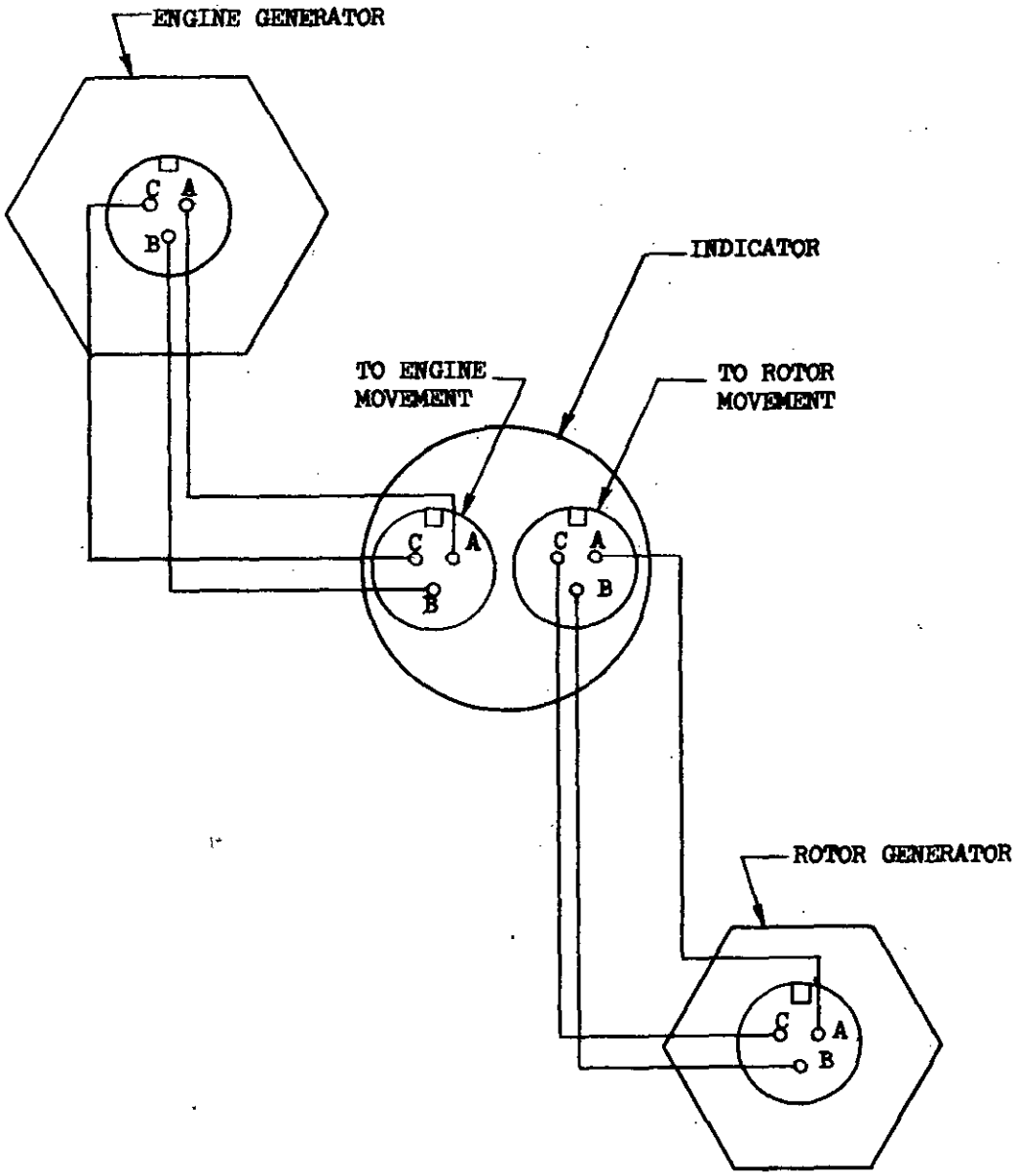
3.6 Dial and pointer -

3.6.1 Dial - The dial shall be securely mounted in such a manner that it will not loosen or slip when the indicator is vibrated. If screws are used, they shall be so located that they will not interfere with the proper location of any dial marking.

3.6.1.1 Visibility of dial - The pointer, numerals, at least 1/3 of the shortest graduations, and all other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 30° with a perpendicular to the dial and whose small diameter is the aperture of the case.

3.6.1.2 Dial marking - The dial shall be marked as shown in Figure 2. All visible portions of the dial shall be lusterless black Color No. 37038 of FED-STD-595 and all markings on the dial shall be lusterless white Color No. 37875 of FED-STD-595. The colors shall not change as a result of service use or subjecting the indicator to the inspections specified herein. The form of the numerals and letters shall be in accordance with MS33558. Numerals shall distinctly indicate the graduation to which each applies. If practicable, each numeral shall be placed so that the center of area of the numeral(s) is on the radial line joining the appropriate graduation and the center of the dial. Any confusion resulting in doubt as to the graduation to which the numeral applies shall be cause for rejection. When several numerals are used in one group, the space between the numerals shall be not less than 1/64 inch.

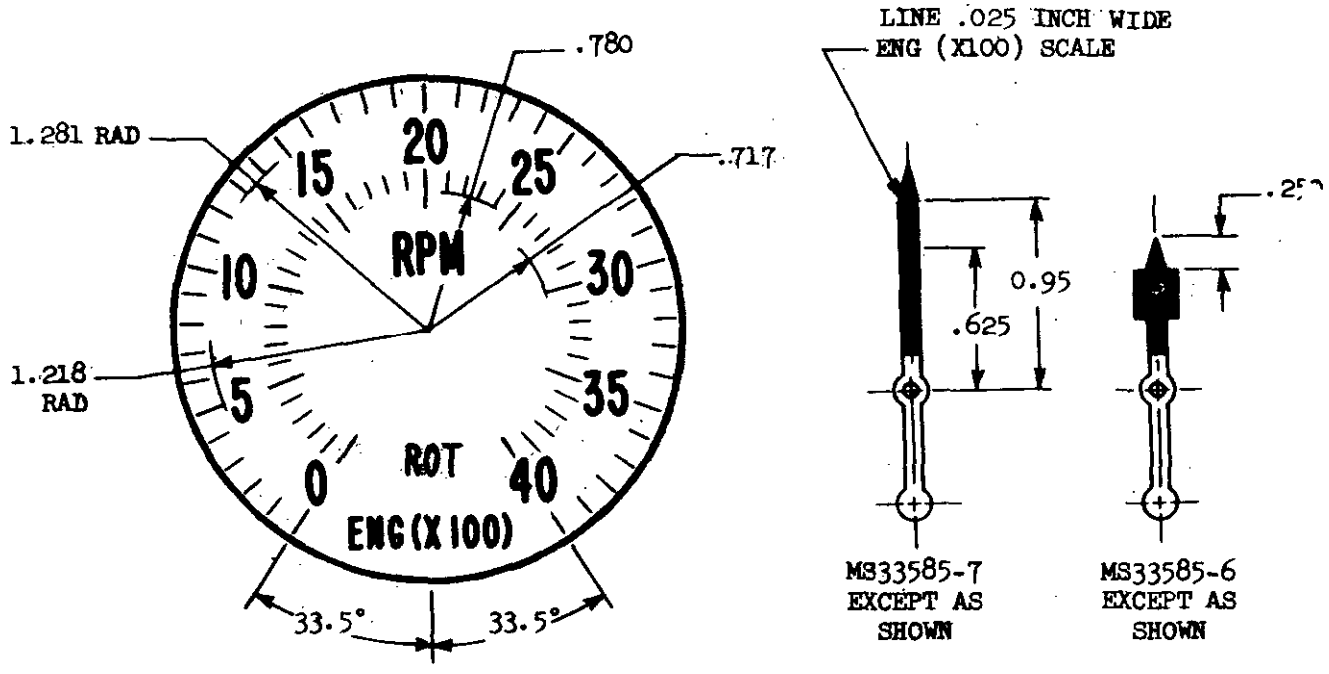
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FOR COUNTERCLOCKWISE ROTATION OF GENERATOR, AS VIEWED FROM SHAFT END.
FOR CLOCKWISE ROTATION, INTERCHANGE A AND B LEADS AT GENERATOR.

FIGURE 1. Connections between indicator and generators

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RPM DIAL

Dimensions in inches, unless otherwise specified tolerances ±0.005 inch, angles ±1/2 degree

Length of respective pointers shall be such that the tip of each pointer shall extend into the respective scale a distance equal to 1/3 to 2/3 of the length of the shortest graduation. The "R" pointer shall be nearest the glass.

MARKINGS	HEIGHT OR LENGTH INCH ±0.015	WIDTH OF LINE OR GRADUATION ±0.005	MATERIAL OR FINISH (See ordering data paragraph 6.2)
NUMERALS 0, 5, 10, 15, 20, 25, 30, 35, 40	0.250	0.031	LUSTERLESS WHITE FED-STD-595 COLOR NO. 37875
SCALE MAJOR GRADUATIONS MINOR GRADUATIONS	 0.219 0.156	 0.025 0.015	
LETTERING RPM ROT and ENG(X100) SHADED PORTIONS OF POINTERS	 0.250 0.188	 0.031 0.031	
BACKGROUND OF DIAL UNSHADED PORTION OF POINTERS	---- ----	---- ----	LUSTERLESS BLACK FED-STD-595 COLOR NO. 37038

FIGURE 2. Dial and pointers

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3.6.2 Pointer - The pointers of the indicator shall be firmly attached to the mechanism but also shall be readily adjustable. The shaded portion of the pointer(s) shall be finished lusterless white, Color No. 37875 of FED-STD-595. The unshaded portion shall be finished lusterless black, Color No. 37038 of FED-STD-595.

3.7 Cover glass - The quality of the cover glass shall be in accordance with DD-G-451, Type II, Quality AA. Any flaws permitted by DD-G-451 shall not interfere with reading of the dial.

3.7.1 Cover glass mounting - The distance from the inner surface of the cover glass to the surface of the dial on which the marking is applied shall not exceed 0.250 inch.

3.8 Performance - The indicator shall perform satisfactorily when subjected to the tests specified in Section 4 of this specification.

3.8.1 Special reliability design - The design of the indicator shall be planned and analyzed as outlined in MIL-R-22256.

3.8.2 Reliability in mean-time-between-failures (MTBF) - The indicator shall have 1000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.9 Identification of product -

3.9.1 Nameplate - A nameplate shall be securely attached to the exterior of the case and shall be marked in accordance with the requirements of MIL-STD-130 except that the FSN shall be omitted, and the manufacturer's part number shall be included.

3.9.2 Manufacturer's part number - The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.10 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-70327.

3.11 Soldering - Soldering shall be performed in accordance with MIL-S-6872.

3.12 Screw threads - Screw threads 0.060 inch in diameter or larger shall be in accordance with MIL-S-7742.

3.13 No data is required by this specification, or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the

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performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection - Inspection of the indicators shall be classified as follows:

(a) Preproduction inspection: Preproduction inspection consists of examination and tests performed after award of contract on sample indicators to determine that the indicator meets the requirements of this specification. The sample shall be representative in design, performance, and configuration of the indicators which will be produced on the manufacturer's production line.

(b) Quality conformance inspection: Quality conformance inspection consists of examination and tests performed on indicators manufactured and submitted for acceptance under contract.

4.3 Preproduction inspection - The preproduction inspection of the indicators shall consist of all of the examinations and tests of this specification performed in the order specified under the paragraph headed Inspection Methods.

4.3.1 Preproduction inspection sample - Preproduction inspection samples shall consist of three indicators manufactured in accordance with this specification. The samples submitted for preproduction inspection shall have been previously subjected only to the Individual inspection. The samples shall be forwarded to the laboratory designated by the procuring activity.

4.3.1.1 Preproduction inspection sample identification - The preproduction inspection samples shall be plainly identified by durable tags, securely attached, and marked with the following information:

Sample for Preproduction Inspection
Indicator, Tachometer, Helicopter, 0-4000 Rt
Submitted by (Manufacturer's name, date) for
Preproduction Inspection in accordance
with MIL-I-8580B under contract or
order number.
Type Number MA-1
Manufacturer's Part Number

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4.4 Quality conformance inspection - The Quality conformance inspection shall consist of the Individual inspection, the Sampling plans, and the Reliability assurance tests of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing the required inspections. Sampling plan B shall be conducted at a Government laboratory designated by the procuring activity. Quality conformance inspection, except for Sampling plan B, shall be under the supervision of the Government quality control representative. The contractor shall furnish test reports showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

4.4.1 Individual inspection - Each indicator submitted for acceptance shall be subjected to the Individual inspection. This inspection shall determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each indicator accepted shall have passed the following inspections:

Examination of product
Scale error at room temperature
Friction error
Position error
Immersion (leakage)
Insulation

4.4.2 Sampling plans - The Sampling plans shall consist of Sampling plan A and Sampling plan B inspections. The inspection samples selected for sampling tests shall first have passed the Individual tests. The test samples which have been subjected to Sampling plan A inspection shall not be delivered on contract until they have been refurbished and resubmitted and passed all the Individual inspections. (See paragraph 6.4) Inspection samples which have been subjected to the Sampling plan B inspection shall not be delivered on contract.

4.4.2.1 Sampling plan A sample selection - Sampling plan A samples shall be selected at random in accordance with the following schedule:

Quantity Offered for Acceptance	Quantity to be Selected for Inspection
First 15	(See Note)
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

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NOTE: When Sampling plan B is invoked, the quantity shall be zero. When Sampling plan B is to be omitted, the quantity shall be one.

When a defective indicator occurs, no items from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. In addition, when a failure occurs, shift to one sample out of the first fifteen (when Sampling plan B is omitted) and proceed as indicated.

4.4.2.1.1 Sampling plan A inspection - Each sample selected for Sampling plan A inspection shall be subjected to the following inspections:

Individual inspection
Low-temperature starting
Scale error at -54°C
Scale error at -35°C
Scale error at 0°C
Scale error at high temperature
Precipitation
Magnetic effect
Vibration error

4.4.2.2 Sampling plan B instructions - Two indicators shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. These samples shall be forwarded to a Government laboratory designated by the procuring activity. Each sample shall be plainly identified by a durable tag, securely attached and marked with the following information:

Indicator, Tachometer, Helicopter, 0-4000 RPM,
Type MA-1
Submitted by (Manufacturer's name, date)
for Production Acceptance Sampling Plan B
inspection, in accordance with Contract/order
No.
Manufacturer's part number _____

4.4.2.2.1 Sampling plan B inspection - Each sample selected for Sampling plan B inspection shall be subjected to the following inspections:

Sampling plan A.
Vibration failure
Extreme low-temperature exposure
High-temperature exposure
Overspeed and reverse operation
Endurance
Thermal shock
Salt spray
Test for helium
Internal examination

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4.4.3 Reliability tests - Test samples which have successfully passed the Individual tests shall be chosen for Reliability Quality Assurance tests. There shall be two kinds of tests:

(a) Predelivery tests - Predelivery tests are tests conducted prior to delivery of initial production indicators or production samples or indicators representative of production samples. The number of samples shall be a minimum of three selected as in the order produced. The maximum number to be used as test samples is at the discretion of the contractor.

(b) Delivery tests - Delivery tests are those tests conducted on indicators starting with initial delivery continuing through the balance of the contract. Samples shall be selected in the order produced. Selection shall be made one out of each fifteen. The contractor may elect to use additional samples, however the number shall not be so large as to adversely affect the delivery requirements of the contract.

All reliability tests shall be conducted by the contractor. A Detailed Reliability Test Procedure Outline shall be submitted to and approved by the procuring activity prior to the initiation of the tests. The test procedure must include, but is not limited to:

(a) Duty cycle.

(b) Performance characteristics to be measured.

(c) Failure criteria.

In any event, the requirements of the following subparagraphs shall constitute minimum procedures and criteria.

Where a contractor is in continuous production from contract to contract he may request a waiver of Predelivery tests if he has already met the requirements for identical items, or similarly identical items. The request shall be additional to, but shall accompany, the above Outline and therefore be subject to approval by the procuring activity.

4.4.3.1 Procedures - The following procedures shall be adhered to for the respective tests:

(a) Predelivery tests.

1. Failure of sample in a way repairable. - The sample may be repaired, or replaced, and the tests continued. The choice is the contractor's.

2. Failure of a sample in a way not repairable. - The sample must be replaced if three only were used initially.

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3. New items placed on test may have design corrections incorporated. This should be noted in the log.

4. To be counted toward the MTBF determination, a total of 3 times the MTBF (as a minimum) must be run on all items.

5. Two samples must demonstrate a MTBF at least equal to the required MTBF. (Not necessarily without failure.)

6. In computing the MTBF, equipments, their equipment running time, and failures may be omitted (censored) provided at least three consecutive equipments of the latest entry are used and all design changes incorporated therein are in the production equipments.

7. All failures shall be analyzed and corrections made in production when appropriate.

8. A log shall be maintained in which pertinent information must be entered. This log shall be available to the procuring activity.

(b) Delivery tests.

1. Failures must be analyzed and production corrections incorporated as applicable.

2. Test results must satisfactorily meet the requirements of the following table:

Failures	Total in Multiples of Specified MTBF
0	4.4
1	5.2
2	6.0
3	6.8
4	7.6
5	8.5
6	9.3

3. All samples must count toward the final MTBF determination unless otherwise agreed to by the procuring activity.

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4. Test results must be recorded and be available to the procuring activity.

5. At least one sample shall run the full required MTBF period, without failure.

4.4.3.2 Test level - The test level shall be III from Table I of MIL-R-23094.

4.4.3.3 Duty cycle - The duty cycle shall be continuous. The duty cycle shall consist of operating the indicator as described in the Endurance test.

4.4.3.4 Performance characteristics to be measured - The scale error at room temperature and Friction error tests, described under Inspection methods, shall be conducted at least once each week. The indicator shall be observed daily, on regular working days, for any obvious failure such as rough pointer movement.

4.4.3.5 Failure criteria - Whenever performance characteristics fall below the acceptance requirements (4.4.3.4) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each should be counted a failure, unless one part caused the other parts to fail.

4.4.3.6 Preventive maintenance - No preventive maintenance shall be performed on the indicators during the Reliability test.

4.4.3.7 Disposition of samples upon completion of tests.- Any sample tested may be delivered on contract provided it meets all the following requirements:

(a) It is representative of production units currently being accepted.

(b) It is in "good as new" condition or has been refurbished. (See pargraph 6.4)

(c) It is otherwise satisfactory.

4.5 Inspection conditions -

4.5.1 Standard conditions - Unless otherwise specified, all inspection required by this specification shall be made under the following conditions:

Temperature	Room ambient, 25° ±5°C.
Pressure	Normal atmospheric (approximately 29.92 inches Hg).

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Humidity	Room ambient up to 90 percent relative humidity.
Input Power	From Generator MIL-G-5413.

4.5.2 Test readings - Unless otherwise specified, before a test reading is taken the indicator shall be vibrated using a vibrator set at 30 cps, 60 cps, or 120 cps with a maximum amplitude of 0.002 inch.

4.5.3 Attitude - Unless otherwise specified, the indicator shall be tested in its normal operating position.

4.6 Inspection methods -

4.6.1 Examination of product - Each indicator shall be examined externally to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests.

4.6.2 Scale error at room temperature - The indicator shall be connected, as shown on Figure 1, to test generators driven at the same speed and tested at all the points of the scale in Table I. The test shall be made by subjecting the generator to the speeds specified, first with speeds increasing, then with speeds decreasing. With speeds increasing, the speed shall be brought up to, but shall not exceed, the speed specified. With speeds decreasing, the speed shall be brought down to, but shall not fall below, the speed specified. The errors at the test points shall not exceed the tolerances specified in Table I. When the generator speed is held constant either pointer oscillation shall not exceed 40 rpm below 500 rpm and 20 rpm above 500 rpm.

4.6.3 Friction error - The generator shall be operated at speeds equivalent to indicated speeds of 1,000, 1,500, 2,000, 2,500, and 3,000 rpm, and the indication before and after tapping per 4.5.2 shall be noted. The speed shall be brought up to, but shall not exceed the speed specified. The indicator shall be in normal upright position, and no reading shall be taken until the pointer has come to rest. The difference in indication before and after tapping shall not exceed 20 rpm.

4.6.4 Position error - The indicator shall be held in the normal operating position, vibrated in accordance with paragraph 4.5.2, and a reading taken when the generator shaft speed is 1000 rpm. This procedure shall be repeated at 45° increments as the indicator is rotated through 360° about each horizontal axis. The maximum deviation of the latter readings from the original reading shall not exceed 20 rpm.

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4.6.5 Immersion (leakage) - The indicator shall be subjected to and meet the requirements of procedure I of MIL-E-5272.

TABLE I

Tolerances

Generator Drive shaft speed rpm	Engine and Rotor Movements					
	Correct Indicated Speed rpm	Tolerances, rpm				
		Room Temp.	-54°C	-35°C	0°C	+71°C
250	500	25				
300	600	25				
400	800	25				
500	1,000*	25	40	40	30	40
600	1,200	25				
700	1,400*	25	45	40	30	40
800	1,600	25				
900	1,800*	25	50	40	30	45
1,000	2,000*	25	50	40	30	45
1,100	2,200*	25	55	45	35	50
1,200	2,400*	25	60	45	35	55
1,300	2,600*	25	60	50	40	55
1,400	2,800	25	65	50	40	60
1,500	3,000*	25	70	50	50	65
1,600	3,200*	30	70	50	50	65
1,700	3,400	30				
1,800	3,600	30	80	50	50	75
1,900	3,800	30				
2,000	4,000*	30	85	55	50	80

* Test points at ambients other than room temperature

4.6.6 Insulation - The stator insulation shall be tested by connecting a test voltage of 200V at a commercial frequency for a minimum period of 5 seconds between any of the connector pins and any metal parts of the case or electrical connector shell. There shall be no breakdown of insulation as a result of this test.

4.6.7 Low-temperature starting - The indicator shall be placed in a chamber, the temperature of which is controllable. The temperature of the circulating air shall be reduced to -54° ±2°C, and held at this temperature for the duration of the test. The indicator shall be kept in the chamber for approximately 2 hours after the temperature -54° ±2°C is reached, before the test is started. No voltage shall be applied to the indicator terminals during this period. Generator speed shall be slowly increased until both rotor and engine movements synchronize. Each movement shall synchronize before 200 rpm generator drive shaft speed.

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4.6.8 Scale error at -54°C. - Following the starting test at low temperature and while still at -54° ±2°C, the engine movement readings at the points indicated by asterisks in table I shall be determined. The errors of the engine unit at any test point shall not exceed the value specified in table I.

4.6.9 Scale error at -35°C. - The indicator readings shall be determined as specified in the Scale error tests at -54°C of 4.6.8, except that the temperature of the air in the chamber shall be -35° ±2°C. The indicator shall be kept at -35°C for at least 2 hours prior to the test. The scale readings shall be determined, starting at 1,000 rpm. The errors of either movement at any point shall not exceed the value specified in table I. The difference between the pointers shall not exceed the width of the band inscribed on the engine pointer.

4.6.10 Scale error at 0°C. - The indicator scale error test shall be repeated at a temperature of 0°C. The errors of either movement at any test point shall not exceed the value specified in table I. The readings of the rotor unit shall not exceed the width of the band inscribed on the engine pointer.

4.6.11 Scale error at high temperature - Each movement scale reading shall be determined as specified in the Scale error at -35°C test of 4.6.9 except that the temperature of the air in the chamber shall be +71° ±2°C. The indicator shall be kept in the chamber for 4 hours after the temperature of +71° ±2°C is reached before the test is started. The errors at any test point shall not exceed the value specified in table I.

4.6.12 Precipitation - Each indicator shall be placed in a chamber, the air temperature of which is controllable. The temperature of this circulating air shall be 85° ±5°C, and the indicator shall be kept in this chamber for a minimum of 2 hours. The indicator shall then be removed from the high-temperature chamber, and within 20 seconds the indicator face shall be immersed to a depth of 1 to 2 inches in a fluid bath maintained at 0° ±7°C. After 3 minutes, the indicator shall be removed from the fluid bath and dried. There shall be no evidence of moisture or other deposits on the inner surface of the glass.

4.6.13 Magnetic effect - The indicator shall be properly connected and operated at 2000 rpm indicated. The indicator shall be revolved about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from the bar magnet. The compass shall have its compensating magnets removed and shall be set up in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19 oersted. The indicator shall be revolved in a horizontal plane which is perpendicular to the axis of the bar magnet. The indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270 and 315 degrees. At each of these positions the indicator shall be rotated 360 degrees about its horizontal axis. The deflection of the compass at any of the specified positions shall not exceed two degrees. This test shall be repeated with the indicator not operating. The same tolerance shall apply.

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4.6.14 Vibration error - The indicator reading shall be determined at any indication between 1,600 and 3,000 rpm, then the indicator shall be subjected to the vibration test, Procedure IV, of MIL-E-5272. The readings with the same generator speed, with vibration shall not vary more than 20 rpm from the readings without vibration. The amplitude of the pointer oscillation shall not exceed 40 rpm.

4.6.15 Vibration failure - The indicator shall be mounted on the vibration stand in its normal operating position. The test shall be conducted in accordance with the Vibration test, Procedure V, of MIL-E-5272 at an indication of approximately 2,000 rpm. After the 3-hour period, the Scale error at room temperature test of 4.6.2 for increasing speeds only shall be conducted. The reading after vibration shall not differ from the reading before vibration by more than 20 rpm. In addition, no looseness in the mechanism nor damage to any part of the indicator shall result from this test.

4.6.16 Extreme low-temperature exposure - The indicator shall be subjected to a temperature of -65° ±2° C for a period of 48 hours. The temperature shall then be raised to -54° ±2° C for a period of 24 hours. The engine movement readings shall be determined as specified in the Scale error test at -54° C of 4.6.8 while still at the -54° ±2° C temperature. The errors of the engine movement at any test point, shall not exceed the value specified in Table I. After the temperature of the chamber has returned to room temperature and the temperature of the indicator has remained at room temperature for a minimum period of 4 hours, the indicator shall be subjected to and meet the requirements of the Scale error at room temperature test of 4.6.2 for increasing speeds only, and the Friction error test of 4.6.3. There shall be no damage which would adversely affect subsequent operation.

4.6.17 High-temperature exposure - The indicator shall be placed in a chamber the temperature of which is controllable. The temperature of the circulating air shall be increased to +71° ±2° C and held at this temperature for a period of 24 hours. At the end of the 24-hour period and while still at the high temperature, the indicator shall operate throughout the entire range without indication of sticking or erratic action. After the temperature of the chamber has returned to room temperature and the temperature of the indicator has remained at room temperature for a minimum period of 4 hours, the indicator shall be subjected to and meet the requirements of the Scale error at room temperature test of 4.6.2 for increasing speeds only, and the Friction error test of 4.6.3. There shall be no damage which would adversely affect subsequent operation.

4.6.18 Overspeed and reverse operation - The indicator shall be connected to a test tachometer generator, and the indicator error determined at generator speeds of 500 and 1500 rpm. The indicator shall be at room temperature, and the errors shall be determined before the indicator has attained an appreciable temperature rise. The indicator shall then be operated, first, at a generator drive shaft speed of approximately 2,200 rpm for a period of 5 minutes, and then in the reverse direction of rotation at a generator drive shaft speed of 1,000 rpm for a period of 1 minute.

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After the indicator has been allowed to cool to room temperature, the indicator errors shall be again determined when the indicator is operated at generator speeds of 500 and 1500 rpm. The difference in readings before and after the overspeed and reverse operation test of 4.6.18 shall not be greater than 25 rpm.

4.6.19 Endurance - The scale errors of the indicator shall be determined when the indicator is operated at generator speeds of 1,000 and 1500 rpm. The indicator shall be at room temperature, and the reading shall be taken before the indicator has attained an appreciable temperature rise. The indicator shall then be operated at an indication of 2,600 ±200 rpm for a period of 1,000 hours. At the end of the 1,000-hour period and after the indicator has been allowed to cool to room temperature, the scale errors of the indicator shall again be determined when the indicator is operated at generator speeds of 1000 and 1500 rpm. The difference of readings obtained before and after this test shall not exceed 30 rpm. No screws or other parts shall become loose or damaged as a result of this test.

4.6.20 Thermal shock - The indicator shall be subjected to 4 cycles of exposure to water at 85° ±2°C and 0° to 7°C without evidence of moisture penetration or damage to coating or enclosure. Each cycle of the test shall consist of completely immersing the indicator in water at 85° ±2°C for a period of 30 minutes, and then within 5 seconds of removal from the bath the indicator shall be immersed for a period of 30 minutes in the other bath maintained at 0°C to 7°C. This cycle shall be repeated continuously, 1 cycle following the other until 4 cycles have been completed. Following this test, the indicator shall be subjected to the Immersion (Leakage) and Precipitation tests of 4.6.5 and 4.6.12, respectively. No indicator leakage or precipitation shall occur as a result of these tests.

4.6.21 Salt spray - One indicator shall be subjected to a salt spray in accordance with Procedure I, MIL-E-5272. External connections shall be made to the indicator in such manner as to simulate installed conditions. Following this, the indicator shall be examined, and there shall be no evidence of external corrosion or deterioration which will affect subsequent operation. The indicator shall then be subjected to the Immersion (leakage) and Insulation tests of 4.6.5 and 4.6.6, respectively, as specified herein. The external surface of the indicator may be washed with distilled water and air-dried prior to conduction of the Insulation test of 4.6.6.

4.6.22 Test for helium - The indicator case shall be punctured, or the filling tube cut, and the indicator subjected to the test for helium by means of a mass spectrometer type helium detector. Failure to detect helium shall be cause for rejection.

4.6.23 Internal examination - The case of the indicator shall be opened and the internal mechanism examined. Any deterioration or damage to the mechanism that resulted from any of the above tests, or any other deficiency, which could in any manner prevent the mechanism from meeting operating or maintenance requirements during service life shall be cause for rejection.

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5. PREPARATION FOR DELIVERY

5.1 Packaging - The indicator shall be packaged in accordance with MIL-P-7936, Level A or C as specified in contract or order (see 6.2). Unless otherwise specified by the procuring activity (see 6.2) for Level A packaging, the method of preservation shall be in accordance with MIL-P-116, Method 1A-5, without preservative compound.

5.2 Packing - The indicator shall be packed in accordance with MIL-P-7936, Level A, B, or C as specified in contract or order (see 6.2).

5.3 Marking - The interior and exterior containers shall be marked as specified in MIL-P-7936.

5.3.1 Precautionary marking - The following precautionary markings shall appear on two opposite sides of each interior package and shipping container wherever practicable.

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended use - The tachometer indicators covered by this specification are intended for use on helicopter aircraft to indicate engine rpm and apparent rotor rpm.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) The quantity, MS part number and type number, if any, of the instrument desired.

(c) Levels of packaging (see 5.1) and packing (see 5.2) desired.

(d) The laboratory that shall conduct tests (see 4.3.1, 4.4, 4.4.2.2).

(e) Method of preservation required if other than method 1A-5.

(f) Items of data required (see 6.3).

6.3 Data - For the information of Contractors and Contracting Officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification, or (c) referenced lower tier documents need not be prepared for the Government and shall not

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be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List) which shall be attached to and made a part of the contract or order. For Navy contracts NavWeps Form 4200/15 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

6.3.1 Reliability - (see 3.8.1, 4.4.3)

6.3.2 Test reports - (see 4.4)

6.4 Definitions -

6.4.1 Amplitude - Whenever the word "amplitude" is used, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.4.2 Hermetic seal - A hermetic seal shall be a perfectly closed and air tight seal made between metallic and metallic or between metallic and vitric materials. A hermetic seal is not intended to include seals which are dependent upon gaskets.

6.4.3 Good as new - "Good as new" shall mean indicators operated less than 10 percent of the specified MTBF operation.

6.4.4 Refurbished - "Refurbished" shall mean that the indicator has been completely overhauled with all component parts meeting current part standards; and the indicator shall have been subjected to and met all the requirements of a new indicator.

6.5 Preproduction test provisions - The manufacture of indicators on contract shall not commence until the samples submitted are pronounced satisfactory by the procuring activity. When a contractor is in continuous production of the indicators from contract to contract, the submission of further preproduction samples on the subsequent contracts may be waived at the discretion of the procuring activity. Approval of preproduction samples or the waiving of preproduction tests does not reduce the requirements for acceptance testing.

6.5.1 It shall be understood that the indicator supplied under contract or order shall be identical to the corresponding preproduction sample in design, construction, quality, material, workmanship, and method of manufacture. Deviation from the standards of the preproduction sample shall be made only by the procuring activity. Evidence of unauthorized change shall constitute cause for rejection.

6.6 The outside margins of this specification have been marked "#" to indicate where changes from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire contract as written irrespective of the marginal notations and relationship to the last previous issue.

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Custodians
Army - MO
Navy - WP
Air Force - No interest

Preparing activity:
Navy - WP
(Project No. 6620-0078)

Review interest:
Army - MO
Air Force - No interest
Navy - WP

NOTICE - Review/user information is current as of date of this document. For future coordination of changes to this document, draft circulation should be based on information in the current Federal Supply Classification Listing of DOD Standardization Documents.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION: MIL-I-8580B INDICATOR, TACHOMETER, HELICOPTER, 0-4000 RPM, TYPE MA-1

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED *(Printed or typed name and activity)*	DATE

DD FORM 1 NOV 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

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NAVY DEPARTMENT

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Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

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